

Omar Alkhadra

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Education

University of Southern California

Master of Science in Applied Data Science (4.0/4.0)

Los Angeles, CA
Aug 2023 – May 2025

George Washington University

Bachelor of Science in Systems Engineering, Minor in Statistics

Washington, D.C.
Aug 2019 – May 2023

Experience

GEICO

Applied Research Scientist, GEICO AI

Data Scientist, Pricing

Chicago, IL / New York, NY

Aug 2026 – Present

Jun 2025 – Aug 2026

- Developed XGBoost and logistic regression models predicting attrition probabilities under rate changes and policy characteristics; $3\times$ reduction in combined calibration error (Brier, ECE, log loss) against the incumbent baseline
- Built stochastic simulation tool with interactive dashboard projecting policyholder attrition under rate-change scenarios, now used to inform pricing decisions across all state markets in production

University of Southern California

Machine Learning Researcher

Los Angeles, CA
Dec 2024 – May 2025

- Co-authored paper published at ACM SIGSPATIAL '25 on probabilistic assignment of noisy GPS trajectories to candidate locations under spatial uncertainty
- Reconstructed proprietary probabilistic baselines (SafeGraph's multi-stage KDE pipeline, Nishida's ILP formulation with spatiotemporal priors) as reproducible benchmarks

AXA XL

Cyber Intern – Data Science

Los Angeles, CA
Jun 2024 – Aug 2024

- Initiated a claim-prediction project with the actuarial team, building multiple ML classifiers on 10,000+ policies to model filing risk from underwriter-assigned categoricals and historical claims data
- Ran multivariate analysis and hypothesis testing to surface the features most associated with claim filing; translated the resulting risk indicators into visualizations for underwriting and product teams

George Washington University

Research Assistant

Washington, D.C.
Aug 2022 – May 2023

- Co-authored peer-reviewed paper in *Environment Systems and Decisions*; built dynamic inoperability input-output model (Leontief) to quantify economic losses across interdependent sectors facing drought
- Conducted sensitivity analysis of policy intervention scenarios with expert-elicited parameters; validated model outputs against published economic loss estimates

Projects

Conditional Calibration in Weather Prediction Markets

- Developed and deployed a systematic strategy on binary weather contracts, trading live against real capital since February 2026 with autonomous daily retraining, execution, and risk controls
- Outperformed market-implied prices on accuracy and calibration: Brier skill +1.2%, expected calibration error 0.010 against the market's 0.030, PIT uniform at KS 0.027 ($p = 0.22$)

Volatility Forecasting and the Statistical-to-Economic Significance Gap

- Built a backtesting framework over five years of SPY options data (2020–2025) with Black-Scholes pricing, regime-conditional variance risk premium estimation, mispricing detection, and daily delta rebalancing
- Diebold-Mariano tests on QLIKE loss showed XGBoost beating GARCH(1,1) ($p = 0.004$, 21% lower RMSE) but neither beating a naive 30-day rolling baseline ($p = 0.26$); hedging methodology, not forecast accuracy, dominated realized returns

Skills

- Programming & ML:** Python (pandas, NumPy, SciPy, XGBoost, scikit-learn, PyTorch), SQL, R
- Tools & Infrastructure:** Cursor, Claude, Git, Linux, Jupyter, Snowflake, Apache Spark, REST APIs, AWS, Docker
- Quantitative:** Probabilistic Calibration, Monte Carlo Methods, Stochastic Modeling, Time Series Analysis

Publications

- Saxena, N., Hsu, S., **Alkhadra, O.**, Shetty, M., Shahabi, C., & Horn, A. L. (2025). POIFormer: A Transformer-Based Framework for Point-of-Interest Attribution. *ACM SIGSPATIAL '25*, 607–619.
- Santos, J. R., **Alkhadra, O.**, Makrides, S., Huang, Y., & Menta, A. (2025). Drought impacts across economic sectors with policy analysis of mitigations. *Environment Systems and Decisions*.